

Teamwork Agreement Template

For the MathWorks Classroom Challenge Projects

Project: Maximizing Solar Panel Output for a Fixed Area

Team Name: SolarMaxxers

Instructor: Dr. Kevin Graham, Dr. Harish Chintakunta, Dr. Esperanza Linares

Program/Class: UCLA EPP

Members and Roles

1. Project Manager (PM) - Maria Dela Cruz
2. Modeling Lead - Sophia Cabalfin
3. Analysis/Validation Lead - Maria Dela Cruz
4. Documentation & Visualization Lead - Ahmed Amin
5. Quality Assurance & Reproducibility Lead - Everyone unless 4th member responds

1) Purpose & Scope

We agree to collaborate to deliver a high-quality, ethical, and reproducible project. Our work will follow the project guidance and produce the specified artifacts (e.g. plots, code, and visualizations).

2) Roles & Responsibilities

- Project Manager (PM): Schedules meetings; maintains timeline; ensures tasks are tracked and disagreements are resolved. → Maria Dela Cruz
- Modeling Lead: Owns core MATLAB models/simulations → Sophia Cabalfin
- Analysis/Validation Lead: Performs hand calculations and cross-checks simulation outputs → Maria Dela Cruz
- Documentation & Visualization Lead: Compiles report (with figures/tables), prepares slides (if needed), and builds visual artifacts (e.g. animations or plots) → Ahmed Amin
- Quality Assurance & Reproducibility Lead: Enforces coherent style across artifacts, ensures work is not overwritten across different versions, ensures data quality, tests/re-runs codes to ensure robustness and reproducibility → Everyone unless 4th member responds

3) Communication Norms

- Primary channel: _Discord_ (e.g. Slack)
- Secondary channel: _____Email_____ (e.g. E-mail for summaries/decisions)
- Response time: ___8 hours_____ on weekdays and **12 hours** on weekends (e.g. within 24 hours on weekdays; within 48 hours on weekends)
- Meeting cadence: _____2 hrs (Friday 6 pm)_____ (e.g. weekly working meeting (60–90 minutes on Tuesdays via Zoom))
- Decision records: ___PM logs major decisions in a running document_____

4) Collaboration & Tools

- Version control: _____ Google Drive folder
_____ (e.g. one shared GitHub repo or Google Drive folder)
- Data management: All datasets, parameters, and assumptions documented in
_____ README.md _____ (e.g. README.md)

File naming & style: _____ descriptive
names _____ (e.g. Snake_case, descriptive names)

- Backups: _____ PM tags old versions according to numbered system _____]

5) Quality Standards (Definition of 'Done')

- Peer reviewed by at least one other member
- Meets all deliverables
- Discussed possible improvements in design as a team
- Code is reproducible (with the [README.md](#) file, specifications, units and assumptions)

e.g. Requirements meet project description, code is reproducible (runs from the README.md file, all specified figures regenerate, at least one teammate has tested edge cases, all assumptions, units, and references are documented in report)

6) Timeline & Milestones

- Week 1: _____ Define objective function and break down
problem _____
- Week 2: _____ Fully implement the Optimization Function

- Week 3: _____ Output of Results

- Week 4 (Due Date: August 6): _____ Final repository cleanup, Interpretation of Results, Report Revision

(e.g.

Week 1: Problem framing, literature/background review, dataset/parameters,

Week 2: First pass models (hand calculations + baseline MATLAB scripts); initial plots/visuals.

Week 3: Validation and quality assurance

Week 4: Report, slides, presentation rehearsal, repository cleanup)

7) Decision-Making & Conflict Resolution

- Default: ____PM facilitate team discussion _____ (e.g. consensus after discussion; PM facilitates
- If blocked >48 hrs: _____ Majority vote _____ (e.g. Time-limited debate, majority vote, PM documents decision)
- Technical disagreements: _____ Team Consensus _____ (e.g. Evidence-based comparison before vote)
- Escalation: If unresolved, consult instructor/mentor with a concise memo.

8) Workload, Accountability & Attendance

- Task board: _____ To Do, File on Google Sheets _____ (e.g. To-Do / Doing / Review / Done with owners and due dates documented in README.md)
- Accountability: _____ Weekly Check-in, In the event of missed deadline, allowed weekend recovery _____ (e.g. Missed deadline → recovery plan within 24 hrs; repeated misses trigger role redistribution.)

- Attendance: _____ Communicate personal-updates through Discord, schedule separate “update meeting” _____
(e.g. If you’ll miss a meeting, give 24-hour notice and share updates asynchronously.)

10) Accessibility & Inclusion

We commit to respectful collaboration, equitable turn-taking, and flexibility for documented accessibility needs.

11) Deliverables Checklist

- ☐ Numerical value for optimal tilt angle
- ☐ Numerical value for optimal aspect ratio
- ☐ Numerical value for maximum energy output
- ☐ Visualization of $E(\theta, r)$
- ☐ Final Report
- ☐ Code (ex. Functions, plots)
- ☐ READ.me file

(e.g. Code repository with clear structure and README, hand calculations linked to modeled results for cross-validation, specific figures & visualizations, final report with specified sections)

12) Agreement & Signatures

Name & Signature: _____ Sophia Cabalfin _____

Date: _____ 7/8/2026 _____

Name & Signature: _____ Ahmed Amin _____

Date: _____ 7/8/2026 _____

Name & Signature: _____ Maria Dela Cruz

_____ Date: __7/8/2026_____

Name & Signature: _____ Date:
